

NIDUS HOME



Why is this project a best practice?

Challenge

The challenge was to turn advanced energy concepts into a full-scale, observable building that could be tested under Romania's continental climate, while remaining understandable to students, professionals and the public. The project also had to coordinate architecture, envelope, renewable systems, controls and user behaviour as one performance system.

Solution

The response combines factory-prefabricated timber-frame modules that shorten site work and improve quality control; natural cellulose-based insulation and careful airtightness detailing for low-energy performance; and a repeatable house concept that can be adapted to site, planning and client requirements while retaining an nZEB-ready envelope. These measures were brought together through an integrated design approach and, where applicable, the Green Homes, nZEB or Passive House performance framework.

Replicability

The project is highly replicable as a method: establish a measurable energy target, prototype the envelope and systems, monitor real operation, and use the results for training and design improvement. Individual technologies can be transferred to schools, offices and homes, although sizing and control must be recalculated for each climate, occupancy pattern and budget.

Key facts and figures

Location	Brezoi, Vâlcea County, Romania
Building use / function	Modular / prefabricated single-family demonstration house
Project type (new build, refurbishment, extension)	New build; modular prototype
Plot area / Floor area	Not publicly disclosed for the registry entry
Year / phase	Prototype / demonstration phase; current product line continues
Climate zone	Humid continental / mountain valley climate
Thematic focus / innovation	Prefabricated timber frame; modularity; natural cellulose insulation; nZEB-ready construction; design for disassembly and repeatability
Investor / building owner	NIDUS MNH / Nidus Home
Architect	Nidus Home / Miradex design team
Other involved stakeholders	Miradex Wooden Buildings; Romanian material suppliers; Romania Green Building Council
Contact person	Nidus Home - https://www.nidushome.com/ ; Miradex - https://www.miradex.ro/
Further information	https://www.rogbc.org/proiecte-certificate-rogbc?lang=en https://www.miradex.ro/ https://en.miradex.ro/catalog-case-lemn-miradex

Nidus Home is a new build; modular prototype project in Brezoi, Vâlcea County, Romania. It functions as modular / prefabricated single-family demonstration house. Its main best-practice contribution is A productised Romanian modular-home concept that links industrialised timber construction with natural insulation and nZEB compliance. The documented strategy brings together factory-prefabricated timber-frame modules that shorten site work and improve quality control, natural cellulose-based insulation and careful airtightness detailing for low-energy performance, and a repeatable house concept that can be adapted to site, planning and client requirements while retaining an nZEB-ready envelope. The public registry and manufacturer information describe the concept; model-specific area and energy values should be requested for each configuration. Public-source research was reviewed in July 2026; data that vary by phase are explicitly flagged in this sheet.

What was done differently?

1. Performance concept - Factory-prefabricated timber-frame modules that shorten site work and improve quality control.
2. Integrated systems and operation - Natural cellulose-based insulation and careful airtightness detailing for low-energy performance.
3. Wider value - A repeatable house concept that can be adapted to site, planning and client requirements while retaining an nzeb-ready envelope.

What was achieved?

Impact

Environmental impact: The project reduces energy and associated emissions primarily by lowering demand and then using efficient or renewable systems. Its material, landscape, waste or mobility measures add benefits beyond operational energy where documented. A

verified whole-life carbon assessment was not publicly available, so the impact is described qualitatively unless a source provides a specific figure.

Economic impact: Lower energy demand can reduce utility exposure and improve long-term operating-cost predictability. Standardised high-performance specifications and third-party certification can also support quality assurance, market value and access to green-finance products. Public sources do not provide a complete investment, payback or life-cycle-cost analysis for this project.

Social impact: The main social benefits are stable indoor temperatures, better air quality, daylight and acoustic comfort, together with safer and more attractive shared or outdoor spaces where these form part of the project. The project also raises market awareness by making high-performance housing visible to residents, designers, contractors and investors.

Key Learnings

Success factors: Key success factors are an integrated design process, a clear performance framework, early coordination of envelope and building services, selection of competent suppliers, quality control during construction and commissioning. Transparent documentation and communication with occupants are essential to preserve the intended performance in use.

Lessons for replication: Replication should start with an energy and comfort target, not a catalogue of technologies. Design teams should verify thermal bridges, airtightness, ventilation, controls and renewable sizing together; record assumptions by project phase; commission systems; and monitor early operation. Where public figures differ by phase, the final as-built dataset must become the single source of truth.